3DU Product Requirements Document

Career discovery and mentorship platform for students in Classes 9–12

1. Purpose and decision context

3DU helps students understand themselves, explore career possibilities and learn from professionals. It brings interests, demonstrated strengths, academic information, family perspectives, career exposure and practical constraints into a coherent guidance experience. The platform helps a student make a better-informed next move without prescribing a final career.

Problem to solve

Students need an understandable path from self-discovery to exploration and reflection. Parents need an appropriate shared view. Mentors need clearly scoped participation. Administrators need traceable evidence, usable review tools and controlled publication. Missing information must remain visible rather than being replaced with plausible answers.

Product goals

- Guide students through nine ordered stages while keeping progress and next actions visible.
- Preserve student and parent perspectives separately and share only the approved six-map report.
- Provide an inspectable four-step mapping workflow, human review and version-specific publication.
- Limit mentors to actual involvement and administrators to authorized records.
- Provide persistent accounts, reliable saved work and accessible browser-based use.

Design principles

Listen before labelling. Explore before deciding. Meet reality before committing. Present personas as provisional summaries rather than diagnoses or permanent identities. Use respectful language, clear status indicators and one visible next action. Routine student and operator tasks should be understandable without developer knowledge.

How to read requirement status

Label	Meaning
Confirmed (C)	A specified product requirement.
Proposed (P)	A concrete experience or workflow proposal that needs product-owner agreement.
TBD / decision (D)	A missing specification, policy or rule that must be resolved before dependent functionality is enabled.
Engineering choice (E)	A reversible technical detail; it does not establish product policy.
Prototype context	Illustrative content; it does not prove implemented behavior or validated findings.

Status applies to the relevant paragraph or table row. Proposed functionality is not a substitute for an approved assessment, completion rule, mapping enumeration or policy. This document is a product review draft; unresolved decisions are listed in Section 9.

2. Users, access and scope

Role contract

Role	Allowed confirmed job	Restricted information/action
Student	Own drafts/submissions, permitted stage review, shared progress, published report, support	No map editing/regeneration, admin/mentor approval, publication, unrelated records or private workflow material
Linked parent	Authorized linked student's identical shared progress and published six-map version, support	No raw assessment answers, private notes, student draft editing, report approval or unrelated students
Involved mentor	Relevant session records and explicitly approved work scope; completion approvals where allowed; initial report review	No map editing/regeneration/publication; no unassigned students or unspecified fields
Admin	Authorized student inspection, evidence correction, mapping inspection, draft editing, requests, reruns, completion capability and version review/publication	No bypass of the initial two-approval gate; no access outside authorized scope

Permission matrix

Capability	Student	Parent	Mentor	Admin
View shared progress	Own	Linked student, same view	Only approved involvement scope	Authorized students
Save stage work	Own and unlocked	No	Session work within approved scope only	Correction method requires defined policy; not impersonated submission
Submit student work	Own and unlocked	No	No impersonation	No impersonation
Approve completion	No	No	Approved stage/student/evidence scope	Approved criteria and authority; exact interpretation remains D
Inspect raw evidence	Own-field visibility requires approved portal design	No	Explicit necessary field scope only	Authorized records
Edit evidence/maps	No	No	No maps; own permitted notes only	Yes, reasons, versions and invalidation
Run mapping	No	No	No	Yes, only current eligible input/specification
Initial report approval	No	No	Involved mentor, after admin, exact version	Exact version and shared-wording review
Later report approval	No	No	Not required	Exact revised version
Publish	No	No	No	Atomic eligibility check and audit
Manage assignments/accounts	No self-link/privilege	No self-link	No self-assignment	P: designated admin capability; approved authorization/verification process required

Relationship requirements

C: student-parent authorization and mentor-student involvement are explicit records, not inferred from an email string or request parameter. C: server checks cover every relevant read/write, including direct URLs and APIs. C: privileged roles are not public signup choices. C: revocation affects access immediately; previously recorded approvals cannot bypass a required current involved reviewer.

P: relationship management uses a browser form restricted to admins with a separately granted management capability, without adding a new public role. Show student, actor, relationship type, permitted scope, verification/reference status, who authorized it, reason and history. Do not collect identity-document values or invent a guardian-verification method. The verification method and who may grant this management capability are D.

P: a designated operator can find unassigned development records through an explicitly approved management view. Ordinary assigned admins do not automatically gain all student access. This proposed capability must not be inferred from a general admin badge.

Scope boundaries

Area	Confirmed scope	Proposed expansion	Decision or dependency
Public website	Explanation, audiences, how it works, start and support	Clear registration/readiness notice and separate admin entry	Approved messaging and launch mode
Identity	Signup/login/logout/recovery; four roles; privileged provisioning	Operator-facing account/assignment forms	Linking verification, identity provider and session policy
Student journey	Nine stages and stateful sequential access	Complete typed forms and source-backed content at each stage	Fields, instruments, content and completion rules
Parents	Linked student’s same progress and published report	Clear link-status explanation	Approved secure linking procedure; no extra questionnaire
Mentors	Relevant records, sessions, completion scope, exact initial review	Work queue and version-specific report review checklist	Approved field scopes and session procedures
Admin	Evidence, four steps, drafts, corrections, reasons, reruns, approvals, publication	Review queue, normal forms, safe assignment/configuration interfaces	Scope of account administration and assignment management
Mapping	Exact four steps and six report	Readable tables, diff previews	Detailed mapping specification

Area	Confirmed scope	Proposed expansion	Decision or dependency
	outputs	and blocker lists	and selected provider
Operations	Persistent storage, instructions and testing	Browser-first operator runbook and readiness dashboard	Hosting, backups, security and policy approvals

Explicitly out of scope

No numerical career-fit scores, alignment calculations, career-cluster rankings, automated mentor matching, required Mindpal connection, embedded Mindpal run URL, invented mentor qualifications, fabricated testimonials/user counts/outcomes, invented pricing, prescribed career decisions or unsupported validated-assessment claims. A parent questionnaire is not added. Payment, subscriptions, analytics, scheduling integrations and automated recommendations are not silently included.

3. Core workflows and functional requirements

Confirmed sequencing

Order	Exact stage	Access
1	Create Profile	Available at start
2	Take Assessments	Stage 1 complete
3	Discover Interests & Strengths	Stages 1–2 complete
4	Explore Career Paths	Stages 1–3 complete
5	Find Relevant Mentors	Stages 1–4 complete
6	Attend Sessions	Stages 1–5 complete
7	Complete Activities	Stages 1–6 complete
8	Get to know yourself	Stages 1–7 complete; publication separately gated
9	Track Progress	Always accessible

C locked message: “This stage will open once you complete the previous stage.” Link to the next available action. Opening, Next, saving or submitting never silently completes work. Completed work remains reviewable. Access checks apply to server/API/direct URLs, not only navigation.

Proposed lifecycle model

State	Meaning	Student action	Authorized review action
Locked	A preceding required stage is incomplete	Go to next available stage; Track Progress	Inspect when authorized; cannot silently bypass sequencing
Available / not started	Accessible stage has no saved work	Open and enter approved fields	No false completion
Draft	Saved, unsubmitted current work	Resume, edit, save, submit if valid	Inspect authorized records; not an approved submission
Submitted / awaiting review	Exact submitted revision awaits applicable review	View submitted copy; proposed withdrawal/edit rule requires owner approval	Review exact revision against criterion
Changes requested	P: reviewer identifies actionable unmet criteria	Read safe feedback and amend permitted work	Preserve prior review; reassess new revision
Complete	Applicable approved completion rule satisfied	Review; continue when sequencing permits	Audited correction/reopening only under approved dependency policy
Setup unavailable	Required instrument/content/criterion not configured	Clear availability message and support	Responsible setup owner resolves; no normal approval button masquerading as available

Locked access is separate from saved/submitted/completion state. These state names are a proposed implementation-facing model; final wording and whether submitted work can be withdrawn are D.

Proposed submission/review contract

Saving stores a persistent draft and confirms the saved time. Submitting creates an immutable reviewable revision with a visible summary. A reviewer sees that exact revision and the approved completion checklist. New changes after submission must either create a new submission or use

the owner-approved withdrawal mechanism; old approvals never transfer. Students see understandable reviewer feedback, not private notes or internal evidence registers. Concurrent edits produce an explicit reload/compare choice rather than overwriting.

Stage rule approval gate

C: completion criteria remain configurable because the source did not supply them. D: admin “may approve at any time” versus evidence-specific authority must be resolved explicitly. P: a stage cannot be enabled for real students until its field/content schema, completion criterion, approver scope, return-for-changes behavior and dependencies are approved as a versioned stage contract. Each submission references that contract version. Configuration changes do not retroactively erase completions without a reviewed migration decision.

P: the setup form asks the owner for the rule in plain language and validates the corresponding implementation config. It must not suggest that filling a reason field alone supplies missing criteria. A completion decision and a review reason are different: the rule specifies what is sufficient; the reason explains why the specific reviewed submission satisfies it.

Each proposal below supplies a concrete experience while keeping unsupplied assessment, activity and mapper content absent. Proposed fields are product choices, not existing confirmed requirements. No stage rule below becomes active through this document alone.

Stage 1 — Create Profile

Confirmed purpose: create/resume a student profile; capture approved relevant background and discovery inputs, with persistent drafts and separate submission/completion.

Proposed screen: short grouped form with an introduction, progress/save state, field-level errors, Save draft and Submit for review. Present only fields approved for this stage; avoid a single generic developer-notes box.

Proposed field group	Candidate fields for owner review	Proposed requiredness	Boundary
Student context	Display name and current class (9–12)	Owner decision	No invented identity/guardian verification; avoid requiring full legal identity merely to explore
Academic context	Relevant subjects and self-described academic context	Owner decision	Do not derive assessment scores or collect school records without approved process
Exploration interests	Interests, activities tried, questions about careers	Proposed optional/free response	Preserve original statement and missing answers; no forced labels
Student goal	What the student hopes to explore next	Proposed optional	Not a promise of a final career decision
Practical context	Approved fields for location/lifestyle/constraints	Defer sensitive scope pending policy	Do not require precise home address, family income or identity documents through an improvised form
Parent involvement	Link-status display only	As authorized	Not a parent questionnaire; linking is separate verified authorization

Proposed review criterion for discussion only: required approved fields are present, the student has submitted, and an authorized admin reviews the profile. Whether completeness alone or substantive evidence is required remains D. A submitted free-text note alone is not an approved completion rule.

Proposed reviewer actions: Approve completion, Request changes, Save review note and Cancel. Show current rule checklist, submitted values, review history and a human-language reason input. Software supplies revision/version IDs. On success show “Profile approved. Take Assessments is now available” only if all relevant rules genuinely pass. If no rule exists, display “Profile completion rule needs setup” with an authorized setup link instead of a misleading normal approval button.

Acceptance proposal: drafts survive refresh/login; incomplete approved required fields prevent submission with field-specific messages; returned work has actionable safe feedback; exact-current-revision approval is enforced; no assessment or report approval is implied.

Stage 2 — Take Assessments

C: no invented validated questions, instruments, scoring rules or career-fit scores. D: instruments, content, licenses, age suitability, versions, response types, scoring interpretation and completion criteria must be supplied.

P experience: an approved assessment catalog explains purpose, source, expected response format and known limitations; supports resume and consent-dependent availability; renders actual approved questions in accessible controls. Show completion/review status rather than a fabricated numerical career-fit result. Ordinary users never see schema JSON. Missing content produces “Assessments are not available yet. Your saved profile is safe.” Detailed missing instrument/version information goes to the setup owner.

P review: show which approved instruments and versions have valid submitted responses and which exact criteria remain unmet. A scoring implementation is enabled only when supplied rules can be reproduced and tested. A manual completion mark cannot falsely certify an assessment was taken. Assessment stages cannot be replaced by invented generic notes for live use.

Stage 3 — Discover Interests & Strengths

C: support exploration of interests/strengths; no invented fit/rank calculations. P experience: show source-backed, plainly worded reflection prompts/content and approved summaries of the student's own inputs. Distinguish a student statement, a reviewer interpretation and an assessment result. Allow the student to identify uncertainties or disagree with a working summary through an approved feedback mechanism. Do not imply scientific validation from a visual label.

D: approved prompt/content inventory, what is displayed from assessments, whether a reflection submission is required, reviewer authority and sufficient evidence. Proposed completion: a defined reflection artifact is submitted and reviewed against an approved checklist, not "opened the page".

Stage 4 — Explore Career Paths

P experience: searchable/browsable approved career information with descriptions of work, learning pathways, practical considerations, source and reviewed date. Student bookmarks or reflections are proposals, not a recommender/ranking system. Any displayed education requirements must be source-backed and context-specific; do not invent employment outcomes, salaries or fit claims.

D: initial content inventory, editorial ownership, filters, field schema and required exploration/reflection evidence. P completion requires the owner-defined exploration activity and submitted reflection if approved; browsing/Next alone never completes. Never make paid-service purchase an unapproved gate.

Stage 5 — Find Relevant Mentors

C: no automated matching; only real authorized mentor data and explicit involvement. P experience: approved mentor directory or admin-assisted request form, using truthful permitted profile information and availability state. A student request is not an assignment or access grant. An authorized operator performs the approved suitability/assignment process; the student sees pending/confirmed status.

D: mentor identity/qualification verification, public profile fields, eligibility, capacity, assignment process, contact boundaries and stage completion. No fabricated affiliations, credentials or implied appointments. Admins see why no mentor is available; initial report review remains pending if no involved mentor can approve.

Stage 6 — Attend Sessions

C: relevant mentor session records and completion capability. P experience: approved session details/status, preparation information, permitted student reflection and attendance/interview review state. Scheduling, rescheduling, meeting links and reminders are separate proposals requiring integration and safeguarding decisions; do not claim a functioning calendar/video integration without one.

D: session format, how attendance/interview evidence is verified, allowed mentor fields, safe communication process, cancellation handling and completion criteria. Mapping requires the approved mentor-interview requirement; a booking or session-page visit is not evidence it occurred. Private mentor notes remain private; shared feedback needs an explicit scope.

Stage 7 — Complete Activities

P experience: source-approved activity catalog with clear purpose, instructions, acceptable evidence and feedback. Support drafts and approved evidence-upload/text mechanisms. Attachment support is only enabled after approved file types, storage, permissions, scanning and retention rules are supplied. Avoid inventing the activities themselves.

D: activity inventory, evidence criteria, required versus optional work, review authority and retry/feedback rules. An uploaded file or clicked button does not demonstrate completion unless the approved criterion says so. Student evidence and reviewer decision remain distinct.

Stage 8 — Get to know yourself

C: unlock follows Stage 7; report publication is a separate gate. Before publication show exactly: "Your mentor and the 3DU team are reviewing your insights. Your results will appear here once approved." Share only the same reviewed six-map version seen by linked parents.

P presentation: six understandable sections with concise explanations, provisional wording and clear published-version/date indicators. Formatting must not expose private confidence, assumptions under review, intermediate classifier rules or evidence registers. Student questions/feedback on a report are a proposal requiring an approved workflow; they do not grant editing or rerunning maps.

D: whether Stage 8 itself has a completion criterion, what approved follow-up reflection occurs, and how this affects progress. Viewing/publishing a report must not silently define an unsupplied Stage 8 completion rule.

Stage 9 — Track Progress

C: always accessible; student and linked parent receive the same authorized shared content. P: show stage name/status, safe saved/submitted/review indicators, the next available student action, report review/published state and support. Parent navigation must not expose the student's raw stage forms. Completion percentages/timestamps are proposed presentation fields; do not imply assessment/career accuracy. Parent/student share the same data content even when role-specific controls differ.

Functional requirements – Required Stage Behaviour

ID	Requirement	Acceptance check
FR-01	Signup/login/logout/recovery/role routing	Applicable acceptance: AUTH and ADM tests
FR-02	Real persistent accounts/data	Applicable acceptance: persistence/resume tests
FR-03	Nine statuses; sequential 1–8; always-open 9	Applicable acceptance: stage direct-route tests

ID	Requirement	Acceptance check
FR-04	Open/save/submit/completion separate; Next not completion	Applicable acceptance: state-transition tests
FR-05	Backend role/stage/direct-URL restrictions	Applicable acceptance: permission matrix tests
FR-06	Draft resume; completed-stage review without silent downstream change	Applicable acceptance: draft/correction tests
FR-07	Exact locked message and next action	Applicable acceptance: UI/API/message tests
FR-08	Unspecified assessments/scoring/activities configurable; no invented validation	Applicable acceptance: unavailable-content/spec tests
FR-09	Admin completion capability; involved mentor scope; relationship to criteria unresolved	Applicable acceptance: configured-authority tests and D-04 decision

4. Mapping workflow and output requirements

Confirmed input coverage

Student/parent responses; educator/mentor notes; academic information; activities/interests; separate student/parent career preferences; strengths, work values, work style; lifestyle preferences; financial/geographical constraints; previous interaction notes. These are supported evidence types, not blanket authorization to collect every field. Collection purpose, sensitivity, consent and requiredness must be approved per field.

Proposed field catalog contract

Each activated field declares identifier, label, purpose, data type/allowed values, requiredness, collection stage/process, source/stakeholder, reader/writer scopes, whether shared in public progress/report, validation rule, specification version and policy dependencies. Sensitive field activation is blocked when its policy dependency is unresolved. Scientific questions/scoring rules are imported from approved sources, not authored by engineering.

Minimum internal evidence structure

Element	Confirmed meaning / proposed refinement
Evidence ID/version	E identifier; immutable attributable versions
Original statement	C retained verbatim; transformations do not overwrite it
Stakeholder	C distinguish student, parent, educator and mentor
Source	C preserve source; P reference origin submission/session/document and author
Evidence/input type	C coverage above; classifier distinguishes fact/preference/behavior/quotation/constraint/expectation/assumption/ contradiction/missing evidence
Career dimension	C classifier dimension; exact taxonomy/import format D
Sentiment/confidence	C internal metadata; taxonomy/calibration D; never a career-fit result
Contradiction status	C identify conflict; detailed taxonomy/resolution process D
Correction lineage	C original plus reason/actor/time; E implementation identifier
Visibility classification	P formal private/shared field scope; D policy and exact field permissions
Missingness	C absent remains absent; P explicit reason only when genuinely known, never guessed

Snapshot/run contract

C: every run references a saved versioned snapshot of its source inputs. P: snapshot also references exact stage submission/criterion versions and the evidence records relied on for readiness. Later editing does not mutate a historical snapshot. Source corrections create a new eligible snapshot and invalidate dependent unpublished outputs. A currentness check covers content and specification versions, not just a timestamp or page refresh.

Record separation

Accounts, relationships, student draft, submitted revision, completion decision, evidence/correction, input snapshot, readiness review, workflow run, individual attempt/output, generated draft, admin-edited draft, report review/approval, publication and audit event are distinct logical records/states. Implementations may choose storage details, but collapsing them must not erase lineage, transfer approvals or overwrite published content.

Readiness

C: mapping follows required assessments, mentor interview and activities. D: exact sufficient evidence, completion rules and reviewing authority. P: a readiness checklist shows each approved prerequisite, evidence/version and unmet dependency. “No mentor available” keeps initial approval pending; it is not a reason to fabricate an interview or bypass review. Required missing parent evidence, where the detailed specification requires it, must remain a visible limitation rather than an invented response.

Step 1 — Evidence Classifier

C: preserve original statement; record stakeholder, source, evidence type, career dimension, sentiment, confidence and contradiction status; distinguish facts, preferences, behavior, quotations, constraints, expectations, assumptions, contradictions and missing evidence. Produce structured JSON internally and a readable evidence register. Never convert an assumption into a fact.

P acceptance: each statement can be traced to the snapshot; unknown source references are rejected; inconsistent attribution is flagged; evidence types are visible in readable columns; the admin can correct interpretation while preserving the original. Classifier output confidence is an uncertainty cue, not scientific certainty. Detailed taxonomies/provider prompts require supplied specification and validation.

Step 2 — Persona Identifier

C: only exact student/parent options from the missing detailed mapping specification. Retain selected hypothesis, alternatives, supporting evidence, needs, barriers, confidence, gaps and assumptions needing validation. Insufficient evidence produces an explicit limitation; do not force a label or stereotype.

P acceptance: option membership is validated against an approved version; parent/student attribution remains separate; no demographic label is treated as sufficient evidence; the reviewer can see why a hypothesis is supported through evidence references and a concise transformation summary. Hidden reasoning is not collected.

Step 3 — Empathy and Need Mapper

C: separate student/parent maps using supplied categories with internal evidence/source/confidence/validation status. Preserve distinctions among warmth, engagement, discovery quality, convenience, parent transparency and student privacy. These distinctions are not a license to invent the missing category enumeration.

P acceptance: required categories match exact approved spec; source references are valid; gaps are explicit; conflict between parent transparency and student privacy is flagged for appropriate human review, not resolved by exposing private student data.

Step 4 — User-Journey Mapper

C: separate student/parent analytical maps with exact eighteen stages and required fields from the detailed specification. Include its specified moments of truth, risks, conflicts, human intervention and automation opportunities. The eighteen analytical stages never control/replace the nine navigation stages.

P acceptance: exactly eighteen approved stages in approved order, all specified fields or honest missingness, validated sources and actor-specific experience. Do not invent a “synthetic production journey” to make output complete.

Exact missing specification delivery

Required source item	What must be supplied	Readiness consequence
Student persona enum	Exact options, definitions, selection/exclusion and insufficiency rules	Step 2 blocked without it
Parent persona enum	Exact options and separate evidence rules	Parent Step 2 blocked without it
Empathy categories	Exact names/order/required fields and student/parent applicability	Step 3 blocked without it
Eighteen journey stages	Exact analytical names/order; not portal labels	Step 4 blocked without it
Journey field schema	Required fields, missingness rules and moments-of-truth/risk/intervention requirements	Step 4 blocked when incomplete
Classification taxonomy	Dimension/sentiment/confidence/contradiction rules and schema	No claimed validated classifier without source
Transformation/public rendering rules	Private versus shared field contract, traceability and review rubric	No public leakage or unsupported rendering assumptions
Specification governance	Source/version, owner, acceptance and change history	No unversioned silently changing mappings

Provider execution and failure

C: private provider credentials/prompts remain server-side; missing provider/specification produces explicit unavailable status, not a fabricated report. Validate structured outputs; provider failure, invalid response and retries are visible. Failed/partial/stale outputs cannot publish and do not damage existing publications.

P: each attempt records input references, spec/provider/config version, status, timing, validation error category and safe error summary. Retry displays affected downstream records and uses idempotent/conflict-safe execution. Retry limits, timeout thresholds, operational ownership and provider selection are D. Logs must not casually copy raw student statements or secrets into public diagnostics.

5. Review, publication and permission rules

Initial report publication

Required work completed → current versioned four-step mapping run → admin review and approval → involved-mentor approval of that same report version → publication to student and authorized linked parent.

Both approvals are required before the first report is published. Admin completion capability does not replace mentor report approval. Admin cannot self-provision a mentor identity merely to bypass the independent requirement. If no involved mentor can approve, show pending; no automatic exception is defined.

Subsequent revision — confirmed

Only admin revises maps. A revised draft needs admin approval of that exact new version before publication. Fresh mentor approval is not required. Retain all earlier published versions/history. Creating a new draft does not remove the last published shared report. Approvals for unchanged historical content remain historical records; they never become approvals of a new version.

Exactly six shared outputs

1. Student persona.
2. Parent persona.
3. Student empathy and need map.
4. Parent empathy and need map.
5. Student journey map.
6. Parent journey map.

C: student and linked parent see the identical approved content version, view-only. Shared text is reviewed for respectful wording suitable to both. Do not expose drafts, partial outputs, prompts, classifier rules, evidence registers, private reasoning, confidence calculations, assumptions under review or private admin/mentor notes.

Proposed publication checklist

Check current authorized publisher; required work/readiness; snapshot currentness; exact mapping spec; four valid current steps; six validated public outputs; no stale dependencies; admin shared-wording review and exact approval; initial involved-mentor approval where applicable; no later published-version regression. The transition and eligibility check are atomic. Repeated clicks yield the same publication result, not duplicate records. A student/parent can never invoke publication through a browser request.

P: preview shows what both audiences will see, without internal evidence. History labels Draft, Needs review, Approved, Awaiting mentor and Published in language suitable to the operator; final labels are owner reviewed. Report downloads/export are proposals, not implied requirements; approved export/privacy rules and accessible document formatting are prerequisites.

Confirmed invariants

- Upstream changes identify and mark dependent outputs as requiring review.
- Affected downstream steps must be rerun before presenting results as current.
- Edited approvals do not carry forward to changed content.
- Draft saving/editing/generation never overwrites the published version.
- Actor, timestamp, exact version, action and relevant reason are recorded.
- Failed runs preserve earlier published reports and history.

Proposed dependency contract

Change	Proposed impact	Operator explanation
Source evidence/submission correction	New snapshot; dependent mapping outputs and unpublished drafts stale	Show before/after, affected run/draft versions, required reruns/reviews
Step 1 output	Steps 1–4 and dependent draft need review/rerun	Classification change can affect every map
Step 2 output	Steps 2–4 and dependent draft need review/rerun	Persona interpretation changes downstream maps
Step 3 output	Steps 3–4 and dependent draft need review/rerun	Empathy change affects journey synthesis
Step 4 output	Step 4 and dependent draft need review	Earlier validated outputs may be reused if current
Public wording edit	New report content version; no inherited approvals	Re-review exact shared wording; initial mentor approval still required before first

Change	Proposed impact	Operator explanation
		publication
Mapping spec/provider-rule change	Determine affected steps through approved dependency rules; new spec/run versions	Do not silently recompute or certify old work
Stage rule change	Owner-approved migration handling for submissions/completions	No retroactive state rewrite through a generic config save
Relationship revocation	Immediate access removal; current approval eligibility checked	No approval bypass by a revoked/uninvolved actor

P: reuse valid upstream outputs only when snapshot/specification dependencies match and lineage is preserved. Final dependency graph and reopening/migration policies are D.

Completed-work corrections

C: reviewing completed work does not silently alter downstream state. P: a correction request previews consequences and asks for an explicit authorized action/reason. Original submissions remain inspectable; student-facing safe feedback distinguishes a correction request from a data-loss event. Whether a completion is revoked, temporarily held or remains completed pending correction is D. Never invent a downstream reset policy.

6. Portals, data, security and quality requirements

Browser-first principles — proposed

Daily work is performed with labelled inputs, clear buttons and readable statuses. IDs, revisions, CSRF values, JSON schemas and execution payloads are handled by software. Advanced diagnostics may offer read-only structured data to specifically authorized technical staff. Editing JSON is never necessary to save a profile, assign a student, review completion, approve a report or publish. A configuration failure must not be presented as an error in the student's work.

Screen inventory

Screen / route concept	Status	Primary action	Required states
Public homepage /	C	Begin or sign in; support; Admin Login	Clear explanation; honest availability; no unsupported claims
Signup /signup	C	Create eligible student/parent account	Validation, existing account, gated collection, success
Login /login	C	Authenticate and route by role	Wrong credentials, expiry, account recovery
Recovery /recover and reset	C	Request/consume supported reset flow	Generic request response, expiry, used token, delivery limitations
Admin Login /admin/login	C	Authenticate admin only	Password visibility, recovery, non-admin denial, expiry
Admin dashboard /admin/dashboard	C	Open authorized student/workflow	Assigned list, empty assignments, permission denial
Student stage screens	C navigation; P detailed forms	Draft, submit, review or next available action	Draft, pending review, complete, locked, source/setup unavailable
Track Progress	C	See shared progress/report	Always available; pending report; published report
Parent portal	C	Shared progress/report	No link, authorized link, pending/published
Mentor portal	C	Relevant session/review action	No involvement/scope, assigned work, pending report review
Submission review queue	P	Select exact pending submission	Empty, filters, stale item, review in progress, missing criterion
Assignment management	P	Authorized audited link/assignment	Missing account, scope validation, verification pending, revoked
Product setup/readiness	P	Resolve precise setup dependency	Missing spec, invalid/incomplete draft, approved active version
Mapping inspector	C; P readable presentation	Inspect/run eligible steps	Pending, valid, stale, invalid, failed, blocked, retry
Report review/editor	C; P form experience	Save revised draft, review exact	Diff, approval status,

Screen / route concept	Status	Primary action	Required states
		version	stale/unpublished, conflict
Publication confirmation/history	C; P checklist presentation	Publish eligible reviewed version	Eligibility failures, prior versions, repeated request

Navigation

C: nine stage labels are exact, with Track Progress always available. C: support persists in authenticated portals. C: public footer exposes clearly labelled Admin Login. P: admin main navigation is Dashboard, Students, Review queue, Mapping, Reports, History and Setup, filtered by permissions. P: daily review screens separate operational tasks from setup/diagnostics.

Public website acceptance

Explain who 3DU serves, how discovery/mentorship/reflection work, student entry, parent/mentor participation and support. Use readable, respectful language. No invented social proof or assured outcomes. Keep personal journey records behind authorization. Proposal: identify whether the site is a fictional demonstration, restricted pilot or approved service before collecting any input.

Authentication and admin access

C: accounts/data persist in real storage; localStorage is not identity or a shared database. Admin Login has email/password, show/hide password, recovery access and server-side role verification; success routes to /admin/dashboard. Anonymous admin page access requires login; signed-in students/parents/mentors are denied. Every admin API requires the relevant role and record scope. Logout and expiry invalidate session use. Privileged provisioning is secure, documented and not exposed as public role selection.

P: authentication forms preserve safe non-password input when validation fails; failures use readable messages without revealing whether a recovery address exists. Show the logged-in account email and role in the account menu so operators do not mistake another admin’s assignments for a defect. Do not display sensitive identity data in public headers.

D: approved session lifetime, email verification, identity provider, recovery delivery channel, operator provisioning verification and account suspension procedures. Live recovery must have an actually working authorized delivery mechanism; a spool-only flow cannot be called live email recovery.

First-admin bootstrap and hosting

C: provide a secure setup command with prompted credentials; no hardcoded password or client-side secret. P: a deployment checklist confirms retained storage, migrations, HTTPS, first-admin access, student assignment, recovery status and required content/rules before calling the site ready. Bootstrap can require a trusted console; ordinary daily operations should not. Restricted bootstrap authorization remains distinct from public registration.

P: document the selected host’s service limits, HTTPS address, renewal and maintenance procedure, and persistence checks. A free hosting option is not an unlimited availability commitment. Public reachability does not authorize live minor-data collection.

Admin portal operating requirements — proposed

Dashboard and student discovery

C: dedicated protected dashboard connects to authorized student records/progress, parent links, mentor involvement, completion approvals, evidence, four-step mapping, draft editing, reruns, approvals, publication and history.

P: show signed-in operator identity, students awaiting authorized review, assigned-student search, missing-setup alerts and tasks awaiting mentor approval. Filters operate only on authorized records. Search results never reveal unrelated student existence. Clearly distinguish “no assigned students”, “no students registered”, “no pending submissions” and “no matching records”. A proposed assignment-management link is shown only to the owner-approved management capability.

Review queue

P: queue items show student, stage, submitted time, applicable criterion version, current review state and permitted reviewer. The operator opens one submission and sees readable values and checklist. Do not require ID copying, revision editing or console commands. Claim/assignment locks and multi-reviewer handling require an approved operating model; regardless, optimistic concurrency must prevent stale review.

P: Request changes includes specific feedback suitable for the student; private operator notes use a visibly distinct input and are never sent by accident. Unsupported delivery channels remain unconfigured. The owner must decide whether in-app feedback alone is sufficient; no implicit email/SMS sending.

Completion review form

UI element	Proposed behavior
Submitted work	Human-readable labelled values; expandable approved attachments; source/version visible to operator without manual entry
Checklist	Approved stage-specific evidence requirements, marked met/unmet/needs clarification based on source-backed review
Review reason	Ordinary labelled text area explaining this decision; not JSON and not a

UI element	Proposed behavior
	replacement for criteria
Approve completion	Enabled only when configured authority/evidence/current revision permit; server independently validates
Request changes	Requires actionable shareable feedback; preserves old submission/review history
Missing criterion	“Setup required” and an authorized setup action; no silent default rule
Conflict	“This submission changed while you were reviewing it. Reload and review the new version.”
Success	Shows completed stage and next genuinely available action; does not approve/publish a report

Evidence editing and requests

C: preserve original statements/source attribution and separate stakeholders; correct evidence with a reason and history. P: evidence register uses labelled columns and source/correction diff. Admin editing cannot silently overwrite the student’s original submission. Requests state the missing information and why it is needed; safe collection/recipient/delivery method must be approved. No improvised parent questionnaire is introduced.

Mapping workspace

C: inspect every step’s input, extracted evidence, transformation summary, output, confidence/gaps, downstream payload and validation/execution state. P: tabs and readable tables show these without forcing raw JSON editing. Advanced structured view is read-only. Run/rerun controls preview affected downstream artifacts and show unavailable prerequisites before execution. No hidden chain-of-thought is requested or displayed.

Report editor and publication

C: only admin edits maps; editing creates a new content version and invalidates inherited approvals. P: a six-section editor shows proposed public wording alongside approved-version comparison, protected evidence links and warnings about private material. Wordings that will be shared with both student/parent are visibly labelled. Exact-version approve/reject/publication actions bind automatically to the reviewed content.

P: publication confirmation shows all eligibility checks, current admin approval, initial involved-mentor approval where required, current run/snapshot/spec versions and safe audience preview. A reason is an ordinary text field. Publish is disabled when ineligible and server-side atomic checks still reject bypass attempts. Repeat requests never duplicate publication.

Product setup workspace

P: authorized setup forms separately cover approved fields/content references, stage rules, reviewer scopes, readiness criteria, mapper spec version, provider/delivery status and policy-release gates. Require draft → validate → owner approve → activate, with actor/time/version/reason. Proposed production configuration approval is separate from report approval. No owner, rule, validated instrument or policy is invented by form defaults. Configuration changes show affected records and require reviewed migration/invalidation actions.

Parent portal

C: show only authorized linked student’s same Track Progress and published six-map report, view-only, with support. No raw answers, private notes, drafts, internal workflow outputs or unrelated students. No extra parent workflow/questionnaire is introduced.

P: if linking is pending/unavailable, explain the status and approved support action without exposing whether an arbitrary student exists. Display which authorized student is being viewed when multiple links are allowed; whether multiple links are supported is D. Never infer a link merely because an entered email matches an account.

Parent evidence remains a supported internal input type, separate from student evidence. D: its approved collection process, attribution/verification and consent. Until decided, preserve missing parent evidence as missing; do not infer a parent persona from a student’s account of a parent as if it were an original parent response.

Mentor portal

C: actual involvement determines access, not role alone. P: show assigned/involved student cards, permitted sessions/tasks and a queue for exact initial report review. Unknown field scope remains restricted. A completion checklist references permitted evidence; if necessary evidence is outside scope, route the task to an authorized reviewer instead of exposing it.

C: mentors cannot edit/regenerate maps. Initial report review shows exactly the admin-approved public version, not a later edited draft. P: Accept/Request clarification are ordinary forms with review reasons; whether mentor rejection returns a draft for admin correction is an owner-approved workflow proposal. Any corrected version requires new exact-version approvals. Session records are attributable to their author; visibility of other mentors’ notes is D, restricted until approved.

P: every enabled assessment, career article, mentor profile, activity, reflection prompt, mapping taxonomy and support message has a source/owner/version and review state. Unapproved content is not used in a live student experience. Superseded content is retained for historical lineage where permitted by policy. Age suitability, localization, accessibility and content-update procedures are D.

C: persistent “Contact for support”; mailto:prashasawan@gmail.com. Phone is labelled “number not supplied / unavailable” and not callable until a real number is authorized. No fabricated support availability or response-time promise.

P: meaningful in-app notifications can indicate submitted, changes requested, approved and published states without leaking private notes. Email/SMS/push delivery and contact-sharing are proposals requiring approved channels, consent, templates and provider configuration. Recording an internal request does not count as successfully delivering a message.

C: no live minor-data collection under invented terms. D: consent, guardian verification, privacy notice, retention, deletion, export, incident process, safety escalation, who can view sensitive evidence and operational owners. Do not assert a jurisdiction-specific age rule or legal compliance without approved advice/policy.

P: a formal release gate separates fictional demo, restricted approved pilot and real service. Each gate lists approved policies, allowed data, accessible accounts/content, integration status and responsible sign-off. Fictional fixtures are labelled and separated from real records. Changing a mode flag alone cannot authorize production collection.

P: field minimization and purpose are reviewed before activation. Sensitive financial/location/family information is collected only when approved and necessary; refusing optional information must have a defined, respectful consequence. Parent transparency never means indiscriminate access to private student evidence. Requests for deletion/export are modelled only after scope, authorization and history/retention treatment are approved.

P: upload/contact/session safeguarding controls must be specified before enabling those capabilities. Malware scanning, attachment permissions, inappropriate-content handling, mentor conduct and incident pathways remain explicit dependencies, not invented operational policies.

Confirmed minimum behavior

Server-enforced role/record/field/stage access; persistent secure accounts; no browser-held provider secrets; protected drafts/internal outputs; reliable state across sessions; versioned migrations; conflict-safe edits; duplicate-safe approvals; atomic publication; clear failure/empty/denied states; responsive keyboard-accessible layouts, labelled forms, visible focus and reduced-motion support.

Proposed nonfunctional acceptance plan

Area	Proposed acceptance method	Target/owner needing approval
Accessibility	Keyboard-only end-to-end tests, labelled errors, focus recovery, screen-reader review, contrast and reduced motion	Conformance standard/level, audit threshold and owner D
Compatibility	Test agreed desktop/mobile browsers and breakpoints on actual devices	Supported browser/device matrix D
Availability/performance	Measure agreed journeys and error rates using approved non-sensitive telemetry	Service targets, traffic/load assumptions and monitoring owner D
Backup/recovery	Restore into separate environment and check accounts/work/publication/history consistency	Frequency, retention, recovery targets and owner D
Security	Threat model, negative authorization tests, credential/session/recovery checks and independent review scope	Security standard, frequency and owner D
Hosting	HTTPS, retained database, migration/runbook, deployment rollback, renewal/service limits	Provider/plan, budget authorization and operational owner D
Integrations	Real provider/delivery contract tests including failure and credential rotation	Selected services, credentials, retry policy and owner D

Do not replace unknown targets with arbitrary numeric success benchmarks. Engineering can measure observed behavior; product/operations must approve required thresholds.

Trigger	Proposed user-visible behavior	Authorized operator detail
Locked stage	Exact confirmed locked message plus next available action	Which preceding completion is unmet
Saved draft	“Your draft is saved” with safe saved state	Persisted revision; no auto-completion
Submission accepted	“Submitted for review” and safe next action	Exact submission/criterion version
Completion rule absent	“This stage is waiting for setup; your work is saved”	Specific missing criterion/authority and responsible setup action
Invalid field	Error beside the field with correction guidance; preserve safe inputs	Validation rule; no raw stack trace
Review reason absent	Labelled reason-field validation	Not “invalid JSON”
Stale review	Explain changed submission and require new review	Current/reviewed versions and diff
No admin assignments	Explain assignment scope; authorized assignment/help action	Distinguish account missing, role wrong and link absent without broad disclosure
Parent not linked	Approved link/support status; no student existence leak	Verification/process record for authorized operator
Mentor scope absent	No raw records; explain pending permitted	Exact approved involvement/field scope

Trigger	Proposed user-visible behavior	Authorized operator detail
	scope	
Mapping spec/provider missing	Clear waiting state; no fabricated report	Exact missing enum/spec/credential destination, excluding secret values
Invalid/provider-failed run	Shared published report remains intact	Attempt, safe failure reason, validated retry controls
Initial mentor unavailable	Pending review; no bypass	Eligible current involvement and same-version approvals
Before first publication	Exact confirmed pending message	Private readiness/review states stay internal
Expired login	Sign in again; avoid falsely claiming unsaved work was saved	Session revocation/expiry checks; preserve unsaved UI text where feasible
Access denied	Understandable denial/support or role portal action	Auditable safe permission event; no private record details
Network save failure	Keep entered work visible and offer retry; no false saved badge	No duplicate submit/publication on retry

Final wording beyond the two confirmed messages requires product review. Diagnostic terms must not overwhelm student/parent flows.

7. User stories and release acceptance

User stories

Story	User value	Testable evidence
US-01	Student saves/resumes	Same draft after refresh/logout and authorized later login
US-02	Student knows available/locked stages	Nine correct statuses; locked URL safe return action
US-03	Parent sees only shared authorized progress/report	Equality of authorized public content; raw/unrelated reads denied
US-04	Involved mentor reviews exact initial report	Role/involvement/version enforced; editing/regeneration denied
US-05	Admin understands mapping	Each step’s source, summary, output, gaps/confidence and downstream status inspectable
US-06	Source correction invalidates dependent outputs	Currentness checks and stale-publication rejection
US-07	Pending results are explained without leaks	Exact pending message; no draft/internal payload exposed
US-08	Revision preserves published version	New draft/version/approval; historical publications retained
US-09	Support is easy to find	Persistent correct mailto and non-callable phone placeholder

Admin authentication acceptance

ID	Status	Contract
ADM-C01	C	Dedicated /admin/login with email/password, visibility control and recovery
ADM-C02	C	Public footer Admin Login link
ADM-C03	C	Persistent authentication; server checks admin role before access
ADM-C04	C	Success redirects to /admin/dashboard
ADM-C05	C	Admin pages/APIs reject unauthenticated/non-admin access appropriately
ADM-C06	C	Secure documented first-admin provisioning; no hardcoded/client credentials or privileged signup
ADM-C07	C	Logout and expiry handling

ID	Status	Contract
ADM-C08	C	Dashboard access to records/progress/relationships/completion/evidence/mapping/drafts/reruns/approval/publication/history
ADM-C09	C	Preserve initial two exact-version approvals and later admin-only approval

Proposed usability acceptance

ID	Status	Proposed acceptance contract
UX-P01	P	Ordinary operational actions never require JSON, IDs or console commands
UX-P02	P	Each enabled stage has an approved typed form/content/completion contract
UX-P03	P	Admin review queue opens exact submissions with readable checklists and safe feedback
UX-P04	P	Missing configuration is distinguishable from student errors; approve controls accurately represent eligibility
UX-P05	P	Authorized assignment management is a browser form with verification/scope/audit checks
UX-P06	P	Account menu identifies signed-in email/role to reduce assignment confusion
UX-P07	P	Shared wording editor/diff/eligibility checklist handles exact version binding automatically
UX-P08	P	Product setup uses labelled versioned forms, with owner approval before activation
UX-P09	P	Readable evidence/mapping inspector; optional restricted read-only structured diagnostics
UX-P10	P	Hosting/bootstrap runbook separate from daily end-user use; no unapproved paid purchase
UX-P11	P	Submitted/reviewed/completed and report-published states are visibly distinct
UX-P12	P	Preserve safe draft inputs on validation/network/conflict errors; do not claim unsaved work is saved

These are acceptance requirements for the agreed release, not a claim they currently pass. Scenario content must use isolated labelled synthetic fixtures until live use is approved.

Test ID	Scenario	Expected outcome
AUTH-01	Eligible fictional student/parent signup	Persistent account; correct portal; no privileged role granting
AUTH-02	Crafted signup/profile request asks for admin/mentor	Server refuses; no privilege change
AUTH-03	Valid/invalid login and role routing	Authorized identity only; wrong credentials handled safely
AUTH-04	Recovery nonexistent/existing account	Same safe request response; delivery actually matches configured mode
AUTH-05	Expired/reused reset token; successful reset	Reject invalid use; successful reset preserves role and revokes sessions
ADM-01	Admin credentials at dedicated login	Dashboard redirect and authorized scope
ADM-02	Anonymous/expired admin page/API	Login requirement/401; no private response
ADM-03	Student/parent/mentor directly requests admin	Server denies, regardless of hidden navigation

Test ID	Scenario	Expected outcome
	page/API	
ADM-04	First-admin bootstrap; duplicated account email	Prompted secret hash; no default credential; no silent role overwrite
ADM-05	Assigned/unassigned admin records	Only authorized student scope; clear correct empty-state meaning
STG-01	Stage 1/Track Progress at start; Stage 2 direct URL	Available/always available; locked message and next action for Stage 2
STG-02	Open, Next, save and submit	No implicit completion or unapproved downstream unlock
STG-03	Save, refresh, logout and later login	Authorized persistent draft restored
STG-04	Missing criterion and otherwise submitted profile	Clear setup-needed status; no fabricated completion or misleading enabled action
STG-05	Valid exact submission/current approved checklist	Allowed reviewer completes; genuinely next stage unlocks
STG-06	Submitted revision changes while reviewer works	Stale review rejected; safe compare/reload prompt
STG-07	Involved mentor lacks required evidence field scope	No leakage or invalid approval; authorized review route
STG-08	Review completed work/correction	Original preserved; approved dependency consequences explicit
STG-09	Admin approves later-stage work outside order under approved rule	Earlier missing completions still gate student navigation
REL-01	Crafted email/student ID link attempt	Does not grant parent or mentor access
REL-02	Authorized assignment/revocation	Audited scope update; immediate access effect
PAR-01	Parent tries raw answers/private notes/another student	Denied; shared report/progress equality remains
MEN-01	Uninvolved mentor; forged student ID; map edit/rerun	Denied server-side
MAP-01	Missing persona/categories/eighteen-stage spec/provider	Honest limitation/blocked generation; no invented values
MAP-02	Snapshot change after run; unknown evidence reference	Output stale/invalid; ineligible publication
MAP-03	Classifier turns an assumption into a fact	Invalid output rejected/flagged; source retained
MAP-04	Insufficient evidence for persona	Explicit limitation and gaps, no forced label
MAP-05	Nine navigation labels substituted for analytical stages	Schema/spec rejection
MAP-06	Step rerun/failure/invalid response	Visible attempt/error; correct downstream invalidation; old publication preserved
REP-01	Initial draft with no/admin-only/wrong-version mentor approvals	Publication rejected
REP-02	Initial exact admin then involved mentor approval	Atomic permitted publication; six validated shared outputs
REP-03	No involved mentor can approve	Pending; no admin bypass
REP-04	Approved draft edited	New version with no inherited approval; previous publication preserved
REP-05	Later revised version with exact admin approval	Publish without fresh mentor approval; retain history
REP-06	Simultaneous/repeated approval/publication requests	Conflict/duplicate-safe, at most one transition per version
REP-07	Stale/partial/failed draft presented as current	Rejected; no public internal/partial output
VIS-01	Student/parent APIs and report UI	Explicit safe schema; identical published version/progress content
UX-01	Daily admin profile review by nontechnical operator	No JSON, revision editing or terminal; checklist and labelled reason sufficient to

Test ID	Scenario	Expected outcome
		operate approved rule
UX-02	Field/reason validation failure	Readable contextual errors, entered safe text retained
UX-03	Keyboard/mobile/reduced-motion journey	Core flows usable, clear focus/labels; approved conformance thresholds checked
OPS-01	Hosted reload/restart and database restore	Persistent accounts/work/reports; no accidental reseeding/reset
OPS-02	Support action/phone placeholder	Exact mailto; no invented callable phone
SAFE-01	Policies absent or mode flag toggled	No authorization to start live minor-data collection

Proposed delivery increments

Increment	Purpose	Entry dependencies	Exit evidence
A — experience/rule agreement	Resolve what the app should do, before further feature coding	Owner review of this draft and missing stage/spec materials	Approved stage contracts, UI scope, decision register and acceptance plan
B — daily-admin usability	Replace JSON/console dependence in approved operations	Approved UX-P scope and assignment-management authority	Browser review/assignment/checklist/draft workflows; negative permission tests
C — real student stages/content	Build approved forms, instruments and content	Approved fields, assessments, career/activity inventory and completion rules	Each stage's functional and source-validity tests; no generic-notes substitution
D — full mapping/report experience	Source-valid generation, readable inspection and publication	Complete exact mapping spec, readiness criteria, provider integration if used	Structured validation, human review, version/invalidation and six-map publication tests
E — approved pilot/live release	Operationally safe hosted use	Policies, guardian process, owners, infrastructure/integrations and final approval	Release checklist, backups/recovery/security/accessibility evidence and explicit sign-off

Delivery increments are proposals; no dates, prices or launch targets are established. Independent work may proceed without treating missing decisions as answered. No static mockup alone counts as functional acceptance.

Required release checklist

The release checklist covers: authentication/recovery/routing; sequencing/direct requests/completion/always-open progress; cross-session persistence; parent/mentor/admin scope/audit; upstream invalidation and failed-run preservation; exact-version initial two approvals/later admin approval; identical shared maps/progress without internal leakage; responsive keyboard/reduced-motion/support; code/schema/setup/verification deliverables.

Proposed readiness checks: all enabled stages have approved content/criteria; daily admin operations are nontechnical forms; unavailable setup is clearly explained; secure linking/provisioning procedures are usable by authorized operators; real delivery/provider integrations work where promised; backup/restore and hosted maintenance are demonstrated; minor-data policies and release approval exist; acceptance distinguishes passed, failed, skipped, blocked and unrun checks.

Definition of done

For an approved feature: source requirement and decision are traceable; typed UI and backend behavior are implemented; permissions, normal/negative/empty/error/concurrency paths pass relevant tests; persisted state survives appropriate restart/login; wording and accessibility are reviewed to approved standards; dependencies genuinely configured; setup/support instructions are usable; no false completion/publication claim exists. A placeholder, hidden button, zero-test run or mocked provider response does not satisfy this definition.

Delivery requirements

Deliver working code, versioned migrations/data models, role/relationship/field enforcement, all agreed screens/forms, real account provisioning/recovery flow, stage contracts/content integration, snapshot/workflow/report/approval/publication lineage, appropriate tests and verification report, variable template without secrets, hosted deployment/maintenance/restore instructions, nontechnical operator guide and missing-decision register. Synthetic fixtures are clearly labelled and isolated.

Before coding a proposed requirement, verify its approval status and supplied materials. Before enabling a stage, confirm its accepted content/fields/criterion/authority. Before generation, confirm exact specification/provider/readiness. Before publication, enforce the required version-

specific approvals. Before live collection, obtain actual policy and release authorization. Continue independent authorized work where another dependency is missing; do not turn a blocked component into fabricated production content.

8. Measurement, dependencies and risks

Success measurement

No numeric targets, baselines, event taxonomy or measurement windows are established. Data collection and measurement require approved privacy controls, definitions and owners.

Outcome area	Definition to agree	Baseline / target
Pipeline progression	Eligible students completing each stage within an agreed cohort and time window	TBD
Progress clarity	Ability to identify the next available action in an approved usability exercise	Method and threshold TBD
Mapping quality	Source traceability, missingness handling, contradiction handling and reviewer corrections by step	Rubric and threshold TBD
Publication integrity	Exact-version approvals, stale-output prevention and preservation of published versions	Monitoring and target TBD
Access control	Unauthorized role, relationship and record reads/writes rejected	Coverage and threshold TBD
Experience health	Authentication, recovery, saving, resuming and workflow reliability	Service targets TBD

Dependencies and blockers

- Approved stage-specific fields, content, completion criteria and reviewer authority.
- Actual validated assessment instruments, permitted use, scoring rules and limitations.
- Approved career information, mentor profiles, session procedures and activity inventory.
- Exact mapping specification: student/parent persona options, empathy categories, eighteen analytical journey stages and all required fields.
- Mapping readiness criteria for assessments, mentor interview and activities.
- Authorized provisioning, relationship verification and mentor field-level access procedures.
- Secure generation and recovery integrations where needed; working persistent hosted storage and maintenance ownership.
- Approved minor-data consent, guardian verification, retention, deletion, export, incident and privacy policies before live collection.
- Accessibility and browser expectations, support ownership and any approved measurement or launch commitments.

Principal risks and controls

Risk	Required control
Unvalidated assessment or invented career-fit output	Use supplied approved instruments only; do not fabricate scores or scientific validation.
Student/parent conflict or unsuitable shared wording	Separate evidence; review shared wording; publish only the approved shared version.
Unauthorized private-data exposure	Server-side role, relationship, stage and record checks; explicit negative tests.
Stale maps after evidence changes	Versioned snapshots, dependency invalidation and blocked stale publication.
Misleading approval controls	Show the missing criterion or authority; do not suggest that a review reason supplies a completion rule.
Unusable routine administration	Proposed labelled forms, review checklists and safe browser operations; technical setup separate.
Failed workflow or conflicting edits	Preserve publications; visible failures; version checks and atomic transitions.
Demonstration mistaken for live readiness	Label synthetic records, disclose unavailable integrations and block unapproved minor-data collection.

9. Open decisions and references

All decisions below remain D unless explicitly approved in a subsequent owner response. Suggested responsible function is a proposal; no person is assigned by this draft.

ID	Decision to resolve	Suggested responsible function	Affected delivery	Precise answer/material needed
D-01	Confirm decision owners/final approver	Product sponsor	All	Names, authority and how approvals are recorded
D-02	Approve proposed nontechnical admin UX	Product/design	Admin experience	Accept/edit UX-P01–12 and representative screens
D-03	Stage 1 actual fields and requiredness	Product/privacy	Student profile	Approved field list, purpose/validation/visibility and optional versus required
D-04	Stage completion versus admin/mentor authority	Product/safeguarding	All stages	Per-stage sufficient evidence, allowed reviewers and out-of-order approval/reopening policy
D-05	Submitted work changes/request-changes lifecycle	Product	Draft/review	Whether withdrawal/editing is allowed; safe feedback and resubmission contract
D-06	Approved assessments/scoring	Assessment owner	Stage 2/mapping	Full instruments/licenses/versions/questions/scoring/limitations/completion
D-07	Interests/strengths reflection content	Content/product	Stage 3	Exact prompts/artifacts, use of results and completion checklist
D-08	Career content and exploration evidence	Content/product	Stage 4	Source inventory, editorial fields/filters/update policy and completion
D-09	Mentor directory/request/assignment process	Mentorship/safeguarding	Stage 5	Real profile fields/verifications/capacity/authority; no automatic matching
D-10	Session/interview/booking evidence	Mentorship/product	Stage 6/readiness	Required session/interview evidence, field scope, cancellation/scheduling and review
D-11	Activity inventory/evidence/review	Content/mentorship	Stage 7	Real activities, evidence rules, required/optional distinctions and reviewer
D-12	Stage 8 completion and follow-up	Product	Stage 8/progress	Whether/how it completes independently of publication; optional feedback/export
D-13	Mapping readiness	Product/mapping	Generation	Required assessment/interview/activity criteria and source-backed checks
D-14	Full detailed mapping specification	Mapping owner	Steps 1–4/six outputs	Exact enums/taxonomies/eighteen names/order/fields and uncertainty rules
D-15	Safe student/parent public wording/schema	Product/privacy	Report sharing	Public field contract, shared wording rubric and conflict/privacy treatment
D-16	Granular dependency/reopening/migration policy	Product/engineering	Edits/reruns	Approved dependency graph, config changes and completed-work consequences
D-17	Assignment/account management authority	Product/security	Admin browser management	Which admins can manage which

ID	Decision to resolve	Suggested responsible function	Affected delivery	Precise answer/material needed
				actors/records and how authority is granted
D-18	Student-parent verification and parent evidence collection	Privacy/safeguarding	Parent links/evidence	Approved verification, collection, revocation and multi-student scope
D-19	Mentor field-level access/notes	Privacy/mentorship	Mentor portal	Exact permitted fields and author/shared-note scope
D-20	AI provider/private prompts and delivery integration	Technical/product	Generation/recovery/ messages	Selected providers, secure bindings, approved source prompts and contract tests
D-21	Minor consent/retention/deletion/ export/incident policies	Privacy/safeguarding	Live collection/release	Approved actual policies, owners and implementation acceptance
D-22	Hosted operations/security/session standards	Technical/operations	Public service	Supported hosting budget, HTTPS/storage/backup/recovery, session policy and security review
D-23	Accessibility/browser/ device expectations	Product/design/QA	Release verification	Approved conformance level/matrix and audit threshold
D-24	Support/notifications	Operations/product	Communication	Real support number or email-only, support owner, approved channels/templates
D-25	Measurement/analytics/ business/launch	Product sponsor	Operational planning	Approved data/event collection, definitions, owners, baselines/targets, budget/model/cohort/date if desired

Approval record template

For each accepted/changed decision record: decision ID; exact approved text/source file/version; approving person and authority; date; affected requirement IDs; rationale; dependent implementation/testing changes; whether it permits fictional demo, pilot or real operation; any remaining limitations. Record explicit rejection/deferment as such. Approving this PRD does not by itself provide the missing instrument/specification files or authorize collection under unapproved policies.

References required

The detailed mapping specification, approved assessment instruments, content inventories and policy documents must be supplied and versioned before dependent features are enabled. Referenced prototype materials are illustrative; no missing specification is assumed to have been read.

PRD owners

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